

Ibrahim Abdul Majeed, PhD

SENIOR PRINCIPAL AI RESEARCHER • ENTERPRISE AI & TECHNOLOGY CONSULTING

Riyadh, Saudi Arabia

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Professional Summary

Senior AI leader with **9+ years of AI/ML industry experience** delivering **enterprise-grade AI solutions** at Samsung, Amazon, and Saudi Arabia's national AI initiatives backed by **PIF and SDAIA**. **Ph.D. in Data Mining & Machine Learning from IISc Bangalore** (M.Tech., IIT Kanpur); **6 US patents** in healthcare AI. Led a **7-member team at Humain** across Agentic AI, Function Calling, and RAG, owning the **full solution lifecycle** from requirements through stakeholder acceptance, and drove the **ALLaM PC product launch** at SDAIA — establishing **organization-wide AI governance and safety frameworks** there. At Samsung, identified **\$500M in potential savings** through a federated learning cost-benefit analysis; at Amazon, **improved customer satisfaction by 10%** on customer-facing conversational AI. Now consulting independently on **AI adoption for KSA government entities** and **FMCG technology**, from client discovery through MVP delivery. Hands-on with **AWS, Azure, and NVIDIA NeMo/Nemotron RL**. **Premium Resident of Saudi Arabia** with an established regional network.

Core Competencies

Enterprise Delivery	End-to-end solution lifecycle: requirements, architecture, development, testing, deployment, stakeholder acceptance
AI/ML Expertise	LLMs, Agentic AI, RAG Systems, Function Calling, Model Fine-tuning & Alignment, Conversational AI, NLP, Healthcare Analytics
Inference Optimization & Serving	Model-level quantization, distillation, KV-cache; serving-level vLLM, SGLang , TGI, Triton with paged & radix attention , continuous batching, and speculative decoding; 60% time-to-first-token reduction
Governance	AI Safety Frameworks, Organizational AI Policy, Cultural AI Safety (Arabic/Islamic Content), Risk Assessment
Leadership	Multi-team Management (7+ reports), Cross-functional Coordination, Client Stakeholder Management, Mentorship
Organizations	Samsung (Global), Amazon (Global), SDAIA (Government), Humain/PIF (Sovereign Fund), NIT Calicut (Academic)
Cloud & Platforms	AWS (multi-company), Azure ML, NVIDIA NeMo/Nemotron RL, VLLM, Amazon Lex & Comprehend, Terraform (IaC)
Rapid Integration	Proven ability to quickly evaluate and integrate third-party libraries, APIs, and cloud services to accelerate solution delivery
Frameworks	Python, TensorFlow, PyTorch, AutoGen, LangGraph, NeMo Aligner, Data Juicer, Transformer Models, CRF
Languages	English (Native), Arabic (Conversational), Malayalam (Native), Hindi (Conversational)

Professional Experience

Confidential / Self-Employed

Independent AI & Technology Consultant

Riyadh, Saudi Arabia

Apr. 2026 – Present

- Advise **government-related entities in KSA** on **AI adoption and digital transformation**, covering AI strategy, architecture standards, vendor evaluation, and implementation oversight.
- Led an **FMCG technology consulting** engagement end to end as a **two-person founding team**: client discovery, **RFP authoring**, and requirements-to-**MVP delivery** of a field-sales platform for 12+ reps, integrated with **NetSuite ERP**.
- Architecting **PlanWise**, a **multi-tenant AI demand-forecasting platform** for FMCG supply chains (statistical baselines, **Light-GBM**, **Temporal Fusion Transformer**), and **LexiMind**, a bilingual Arabic/English **legal AI assistant** built on a multi-agent **Lang-Graph** pipeline with **hybrid RAG**.

Humain (PIF-supported AI company)

Senior Principal AI Researcher

Riyadh, Saudi Arabia

Apr. 2025 – Mar. 2026

- Led a team of 7 across Agentic AI, Function Calling, and RAG teams, **managing the full solution lifecycle** from requirements through deployment and stakeholder acceptance (reporting to Head of Data and Modelling).
- Drove PIF's strategic initiative for **enterprise-grade agent-based AI systems**, coordinating cross-functional teams to deliver unified, deployable solutions.
- Managed **comprehensive project delivery** including model development, integration testing, user acceptance testing, and production handover for ALLaM's agentic capabilities.
- Led **culturally-sensitive AI content verification** for Arabic LLM deployment—ensuring accuracy for religious and regulatory content critical to KSA government and enterprise clients.
- Leveraged **Azure cloud infrastructure and NVIDIA NeMo/Nemotron RL platforms** for large-scale model training and deployment.
- Rapidly evaluated and integrated **third-party libraries, APIs, and cloud services** to accelerate development cycles and deliver solutions ahead of schedule.
- Established **technical roadmaps and strategic direction** for advanced AI research initiatives across multiple product lines.
- Delivered **35% improvement in model convergence** through curriculum-based training pipeline, directly reducing iteration cycles for downstream solution delivery.
- Optimized **LLM inference serving** — continuous batching, KV-cache, and streaming — **reducing time-to-first-token by 60%** while maintaining accuracy, improving end-user experience for production deployments.

National Center for AI, KSA (under SDAIA)

Riyadh, Saudi Arabia

Expert Researcher (Senior Lead, ALLaM LLM & Agentic AI Team)

Jul. 2023 – Mar. 2025

- Served as senior lead for Saudi Arabia's national AI model (ALLaM), responsible for **delivering production-ready AI solutions for government applications**.
- Delivered AI capabilities for the **ALLaM PC product launch**, coordinating model training, integration testing, and user training across multiple teams.
- Contributed to **LLM tool calling integration** for government applications including Tawakkalnaa, supporting initial development and prototyping phases.
- Delivered **enterprise-grade AI solutions** for meeting scheduling, email drafting, and inbox summarization—functional productivity tools for the ALLaM PC platform.
- Built and delivered a **functional financial assistant agent** with custom-trained models, demonstrating end-to-end practical AI applications with user interface integration.
- Developed **organization-wide AI safety and governance framework** under the Chief Research Scientist, including policy documents, implementation guidelines, compliance monitoring, and adoption strategy.
- Contributed to the **Baseer project for the Holy Mosque**—a nationally significant initiative requiring stringent quality and compliance standards.
- Led **LLM evaluation and benchmarking** providing structured analysis frameworks for model selection, performance optimization, and go/no-go deployment decisions.
- Managed **end-to-end data pipelines** including data curation, preprocessing, quality assurance, and integration with downstream production systems.
- Delivered **technical presentations at SDAIA's AI Winter School** (December 2024) to enterprise and government stakeholders.

Amazon India (International Machine Learning Group)

Bangalore, India

Applied Scientist II

Apr. 2022 – May 2023

- Delivered **customer-facing intelligent chatbot system** providing purchase guidance for high-value electronics on Amazon's global e-commerce platform.
- Achieved **10% improvement in customer satisfaction metrics** during product trial, demonstrating measurable business impact.
- Collaborated with **global Amazon teams across multiple geographies** to create an offline store-like customer guidance experience, managing cross-team dependencies and integration.
- Built on **AWS cloud infrastructure** using Amazon Lex, Comprehend, and related NLP services for production-grade conversational AI with named entity recognition, slot filling, and transformer-based intent classification.

Samsung R&D Institute India - Bangalore (SRIB)

Bangalore, India

Senior Chief Engineer

May 2017 – Mar. 2022

- **Enterprise-Wide Strategic Assessment:**
 - Conducted **comprehensive cost-benefit analysis of federated learning** across Samsung's consumer electronics ecosystem, identifying approximately **\$500M in potential savings** and influencing strategic technology investment decisions.
 - Delivered **feasibility study with implementation roadmap** covering technical assessment, business case, and organizational readiness for privacy-preserving AI at Samsung scale.
 - Research outcomes directly **informed enterprise-level technology strategy** for Samsung's approach to privacy-preserving machine learning in consumer applications.
- **Healthcare AI Solutions (6 US Patents):**
 - Developed glucose prediction algorithms achieving **Region A clinical accuracy** (within 20% of reference on Clarke Error Grid), delivering complete solutions from algorithm development through validation and integration testing.
 - Contributed to **six US patent filings** in healthcare AI, spanning non-invasive monitoring techniques, signal processing, and spectroscopy methods.
 - Collaborated with **Samsung Advanced Institute of Technology (SAIT)** on cross-organizational research initiatives for healthcare solution development.

National Institute of Technology, Calicut

Assistant Professor

Calicut, India

Aug. 2015 – May 2017

- Taught Pattern Recognition, Data Mining, Data Structures & Algorithms, and NLP. Mentored student research projects and maintained active industry collaborations.

Emuzed India Pvt Ltd (Flextronics Software)

Software Engineer

India

Mar. 2005 – Jul. 2006

- Developed multimedia codec plugin solutions for Nokia N series phones.

HCL Technologies Ltd.

Member Technical Staff

India

Jul. 2004 – Nov. 2004

- Early career technical staff role in software development.

Education

Indian Institute of Science (IISc), Bangalore

Ph.D. in Data Mining & Machine Learning

Bangalore, India

Aug. 2009 – May 2016

- Thesis: "Effective Characterization of Sequence Data through Frequent Episodes"
- Advisor: Prof. P. S. Sastry
- Coursework CGPA: 6.3/8
- Relevant courses: Pattern Recognition and Neural Networks, Information Retrieval, Stochastic Models and Applications, Linear and Non-Linear Optimization, Real Analysis

Indian Institute of Technology (IIT), Kanpur

M.Tech. in Computer Science and Engineering

Kanpur, India

Completed Jul. 2008

- Thesis: "Shortest Path in Multi Objective Environment"
- Advisor: Prof. Shashank K Mehta
- CGPA: 8.66/10
- Relevant courses: Randomized Algorithms, Data Streaming Algorithms, Bioinformatics, Computational Geometry, Storage Computing Architecture, Wireless Networks

National Institute of Technology (NIT), Calicut

B.Tech. in Computer Science and Engineering

Calicut, India

Completed Jun. 2004

- Percentage: 73.4%

Selected Publications & Patents

JOURNAL PUBLICATIONS

Discovering Compressing Serial Episodes from Event Sequences

Knowledge and Information

Systems, 47(2):405-432

(Springer)

Ibrahim A., Shivakumar Sastry, and P. S. Sastry

2016

Pattern-growth based frequent serial episode discovery

Data & Knowledge

Engineering, 87:91-108

(Elsevier)

Avinash Achar, Ibrahim A., P.S. Sastry

2013

CONFERENCE PUBLICATIONS

Factors Influencing Cost and Performance of Federated and Centralized Machine Learning

Ibrahim Abdul Majeed, Hwang-Ki Min, et al.

IEEE 19th India Council
International Conference
(INDICON)

2022

Motion Artifact Removal of Photoplethysmogram (PPG) Signal

Ibrahim Abdul Majeed, Sujit Jos, Rahul Arora, et al.

IEEE Engineering in Medicine
and Biology Conference (EMBC)

2019

Analyzing Similarities of Datasets using a Pattern Set Kernel

Ibrahim A., P. S. Sastry, and Shivakumar Sastry

Pacific-Asia Conference on
Knowledge Discovery and Data
Mining, pp. 265-276

2016

PATENTS

6 US patents in healthcare AI, filed 2019–2020 at Samsung R&D — non-invasive glucose monitoring via **near-infrared and diffuse-reflectance spectroscopy**, **PPG signal** pre-processing, and drift, noise, and motion-artifact correction. Representative filings: US 20210396662A1, US 20200237312A1, US 20200352517A1. Full list available on request.

Languages

Languages English (Native), Malayalam (Native), Arabic (Conversational), Hindi (Conversational)

References

References available upon request.